



Sources and timing of pyroxenite formation in the sub-arc mantle: Case study of the Cabo Ortegal Complex, Spain



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ABSTRACT

Pyroxenites exposed in ophiolites and orogenic peridotite massifs may record petrogenetic processes occurring in mantle domains generated and/or transferred in supra-subduction environments. However, the timing of their formation and the geochemical characteristics of their source region commonly are obscured by metamorphic and metasomatic overprints. This is especially critical in arc-related environments, where pyroxenites may be formed during the differentiation of primitive magmas. Our approach combines Sr- and Nd-isotope geochemistry and geochronology, and modelling of REE diffusion, to further constrain the origin of a well-characterized set of pyroxenites from the arc-related Cabo Ortegal Complex, Spain. In the light of petrological constraints, Sr- and Nd-isotope systematics consistently indicate that cpx and amphibole have acquired disequilibrium during two main episodes: (1) a magmatic/metasomatic episode that led to the formation of the pyroxenites, coeval with that of Cabo Ortegal granulites and corresponding to the incipient stage of a potential Cadomian arc (459–762 Ma; isochron and second-stage Nd model ages); (2) an episode of metamorphic amphibolitization upon the percolation of relatively unradiogenic and LREE-enriched hydrous fluids, subsequent to the delamination of the pyroxenites from their arc-root settings during Devonian subduction. Calculations of diffusional timescale for the re-equilibration of REE are consistent with this scenario but provide only poor additional constraints due to the sensitivity of this method to grain size and sub-solidus temperature. We thus emphasize the necessity to combine isochron ages and Nd model ages corrected for radiogenic ingrowth to put time constraints on the formation of subduction- and/or collision-related pyroxenites, along with petrological and geochemical constraints. Homogeneous age-corrected $^{143}\text{Nd}/^{144}\text{Nd}$ of 0.5121–0.5125 (ϵ_{Nd} between 0 and +7.5) and $^{87}\text{Sr}/^{86}\text{Sr}$ of 0.7037–0.7048 provide information on the sources of the metasomatic agents involved (and potentially the parental melts) and notably indicate the contributions from enriched mantle components (EM I and/or II). This suggests the involvement of an old crustal component, which is consistent with the derivation of the pyroxenites and granulites from an ensialic island arc, potentially built on the northern margin of either Gondwana or a pre-Gondwanan continental block. This case study thus documents the role of melt–rock reactions as major pyroxenite-forming processes in the sub-arc mantle, providing further constraints on their sources and timing in the Cabo Ortegal Complex.

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1. Introduction

The mineralogical and geochemical diversity of pyroxenites observed in peridotite massifs and xenoliths provides a unique record of the petrological processes contributing to the evolution of lithospheric mantle domains (Downes, 2007, for a review).

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Despite their minor abundance, their geodynamic significance is strengthened by their implication in the petrogenesis and transfer of mantle-derived magmas in various tectonic environments (e.g. Sobolev et al., 2007). Different types of mantle pyroxenites have been reported, representing crystallized products of *in-situ* partial melting or metamorphic segregations of peridotites (e.g. Dick and Sinton, 1979), metamorphosed products of recycled oceanic lithosphere (Allègre and Turcotte, 1986) and high-pressure crystal cumulates from migrating mantle melts (e.g. Bodinier et al., 1987). Studies of exhumed mantle terranes, such as Ronda (Spain),

Beni Bousera (Morocco), Lherz (France), Balmuccia (Italy), have increasingly emphasized the role of melt–peridotite interaction (Rivalenti et al., 1995; Lenoir et al., 2001; Le Roux et al., 2007; Gysi et al., 2011), among the wide range of mechanisms responsible for the formation of pyroxenites and their compositional diversity.

In the sub-arc mantle, such as that exposed in the Kohistan complex (Pakistan), the formation of pyroxenite layering has been especially ascribed to melt infiltration and melt–rock interaction (e.g. Burg et al., 1998), and it is increasingly accepted that pyroxenite genesis significantly contributes to the differentiation of primitive arc magmas (Conrad and Kay, 1984; Lee et al., 2006). While these melt–rock interaction processes (e.g. crystal segregation in conduits, peridotite and pyroxenite replacement, partial melting) are relatively well understood, they remain in many cases poorly correlated to regional tectonic and geodynamic events such as asthenospheric upwelling, subduction and exhumation (e.g. France et al., 2015; Le Roux et al., 2016). In addition, the comprehensive understanding of their role in the compositional evolution of arc magmatism is commonly hampered by the lack of crystal-fractionation linkages between the residual mantle and igneous crust (e.g. Greene et al., 2006; Garrido et al., 2007). It is therefore pertinent to aim for better chronological constraints on pyroxenite-forming melt–rock reactions and identification of the source(s) of melts (and fluids) involved.

We here present a case study of a pyroxenite-rich sub-arc mantle domain. The Herbeira massif of the Cabo Ortegal Complex exposes a kilometre-sized body with individual pyroxenite layers up to 3 m-thick (Girardeau et al., 1989). They represent cumulates from primitive arc melts of picritic to boninitic affinities that reacted with their host peridotites (Gravestock, 1992; Tilhac et al., 2016; Tilhac, 2017). However, the timing of their formation and the nature of the melt sources are poorly constrained (Santos Zalduegui et al., 1996; Santos et al., 2002) despite the availability of a robust tectono-metamorphic framework (e.g. Ábalos et al., 2003). The present study combines Rb–Sr and Sm–Nd geochemistry and geochronology for a suite of well-characterized pyroxenites along with some constraints from diffusional timescale calculations based on rare earth elements (REE) thermometry (Liang et al., 2013; Liang, 2014). The aims are to (1) provide isotopic constraints on the petrogenesis of Cabo Ortegal pyroxenites, (2) identify the potential source(s) of melts and fluids and (3) add timing constraints to their formation and evolution with respect to regional tectonics, in order to (4) discuss the source and timing of pyroxenite-forming processes in the sub-arc mantle.

2. Geological context

The Cabo Ortegal Complex is part of the Variscan belt of Europe, which comprises five allochthonous complexes in north-western Iberia (Fig. 1). Their subdivisions are correlated into structurally distinct units of high-pressure and intermediate-pressure metamorphic rocks and ophiolites that record peri-Gondwanan arc magmatism and the Gondwanan–Laurussian convergence following the closure of oceanic domains, potentially including the Rheic Ocean (Arenas et al., 1986; Santos Zalduegui et al., 1996; Martínez Catalán et al., 2009; Albert et al., 2015).

In the Cabo Ortegal Complex, mafic and ultramafic mantle rocks are exposed in three massifs (Herbeira, Limo and Uzal; Fig. 2a) along with high-pressure gneisses, eclogites and granulites, within the high-pressure, high-temperature (HP–HT) units of the Upper Allochthon (e.g. Arenas et al., 1986). The Herbeira massif is the largest exposure (~15.5 km²), consisting of a dominantly harzburgitic plateau to the east and lithologically heterogeneous cliffs to the west (Fig. 2b). The latter preserve abundant pyroxenites as-

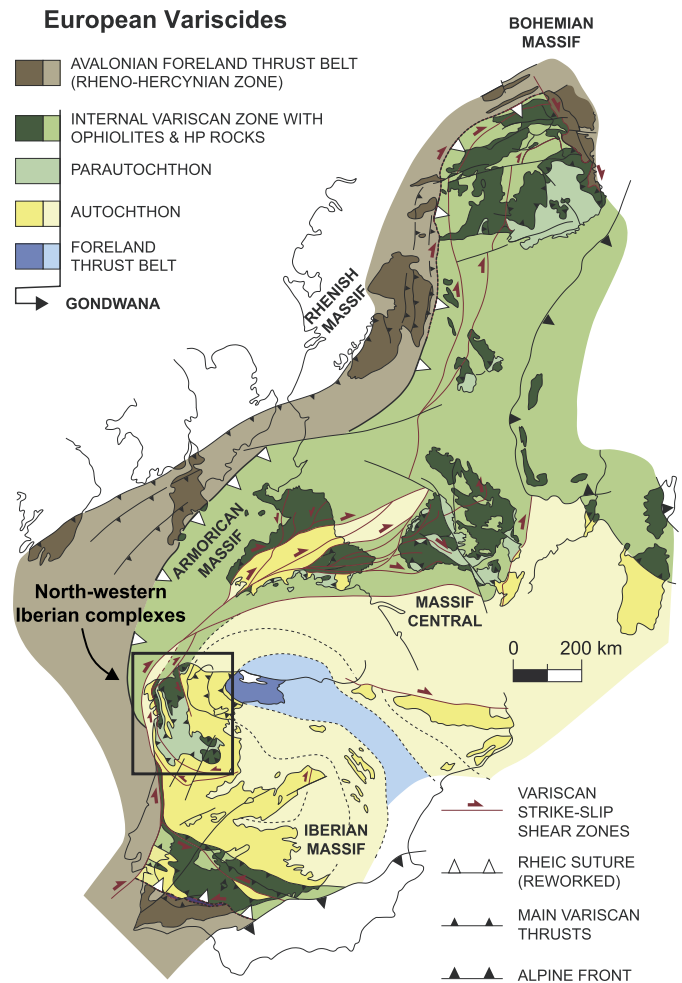


Fig. 1. Location of the north-western Iberian complexes and other European Variscides during the Palaeozoic convergence (Diez Fernández and Arenas, 2015).

sociated with dunites ($\pm$ chromitites) and locally with harzburgites (Girardeau et al., 1989; Girardeau and Gil Ibarguchi, 1991), comprising up to 90% of a ~3 km-long, 300 m-thick domain of one or several lenticular bodies (Tilhac et al., 2016). Pyroxenites and peridotites exhibit HT plastic deformation features including a tectonic foliation and boudinage sub-parallel to the compositional layering. In addition, sheath folds, similar to those reported in the Limo massif (Puelles et al., 2012), and locally mylonites, are observed in the lower part of the cliffs affecting both peridotites and pyroxenites (Tilhac, 2017). This provides a robust constraint of the genesis of pyroxenites pre-dating the onset of high-shear strain deformation (i.e. the development of sheath folds) ascribed to subduction channel processes (Puelles et al., 2012).

3. Material and methods

3.1. Sample collection

24 samples of pyroxenite (and two peridotites) collected in the Herbeira massif have been selected as representative of the four types of Cabo Ortegal pyroxenites defined petrologically by Tilhac et al. (2016) and Tilhac (2017). These different types have distinctive geochemical characteristics and petrogenetic interpretations. Type-1 pyroxenites are dominantly olivine-bearing clinopyroxenites sampled from pyroxenite layers that enclose dunite lenses,

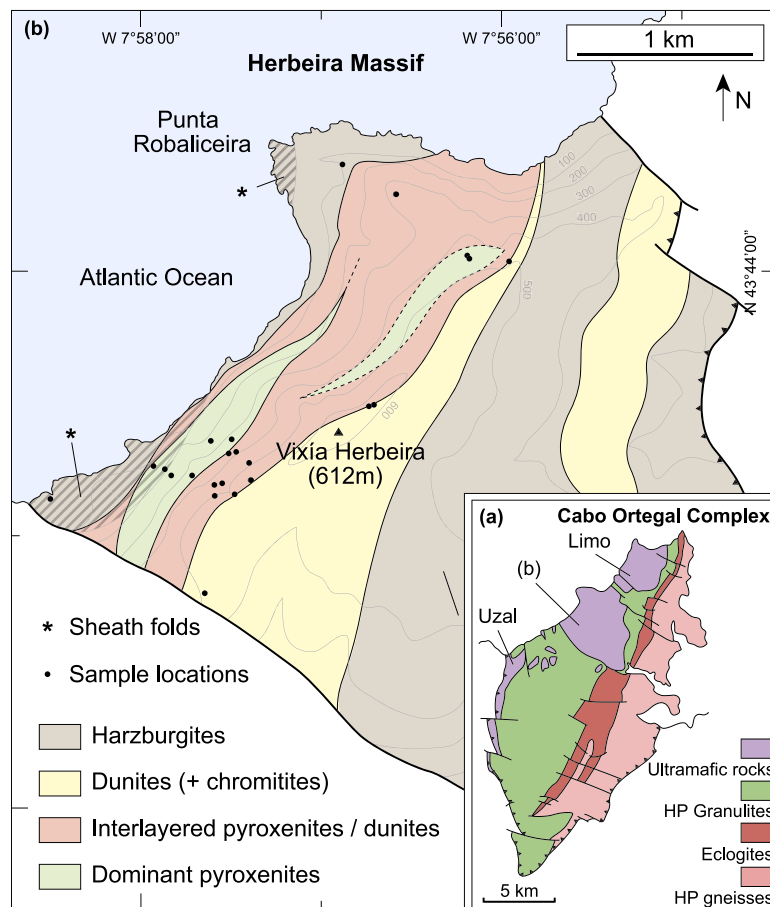


Fig. 2. (a) Details of the high-pressure units of the Upper Allochthon in the Cabo Ortegal Complex and (b) geological map of the Herbeira massif showing sample locations across the pyroxenite-rich domain (after Tilhac et al., 2016; and references therein).

characterized by high Mg# and Cr₂O₃, low Al₂O₃ and TiO₂ and spoon-shaped REE patterns. They are interpreted as the products of partial replacement of the host peridotites upon melt–rock interaction. Type-2 pyroxenites are massive, olivine-free websterites with lower Mg# and notably higher Al₂O₃ and exhibit a wide range of LREE enrichment. They represent either the products of similar melt–rock interaction at higher melt/rock ratios or crystal cumulates from more evolved melts intruded as dykes or veins. Most type-3 pyroxenites are foliated clinopyroxenites, commonly highly amphibolitized, although all pyroxenites are amphibolitized to some extent (5–38 wt%). They have a range of Mg# overlapping that of type-1 and -2 pyroxenites but lower Al₂O₃ at a given Mg#, and exhibit homogeneous LREE enrichment suggesting a chemical equilibrium with their parental melt and/or metasomatic agent. Tilhac (2017) and Henry et al. (2017) interpreted this type as the products of deformation and hydration of a protolith similar to that of type-1 pyroxenites. Type-4 pyroxenites are subordinate opx-rich websterites and orthopyroxenites with high Mg# and Cr#, probably produced from the interaction between Si-rich melt and dunites following an olivine-consuming (i.e. peritectic) reaction (Tilhac et al., 2016). Cabo Ortegal pyroxenites have granoblastic (i.e. equigranular) to porphyroclastic (i.e. with a bi-modal grain-size distribution) textures, where amphibolitization has variously overprinted deformation-related textures (Henry et al., 2017; Tilhac, 2017). Detailed descriptions of textural relationships are given by Girardeau and Gil Ibarguchi (1991), Gravestock (1992), Tilhac et al. (2016), Tilhac (2017).

3.2. Sr and Nd isotopes

Sr- and Nd-isotope compositions were measured on cpx and amphibole separates (and one garnet separate) and whole-rock powders of pyroxenite (and two whole-rock samples of peridotite) in the Geochemical Analysis Unit (GAU) of GEMOC/CCFS at Macquarie University, Australia. The same portions of sample used for the whole-rock elemental analyses reported by Tilhac et al. (2016) were disaggregated using high-voltage pulses in a selfFrag® apparatus and sieved into three fractions (<100 μm, 100–600 μm and >600 μm). Clean mineral separates were then hand-picked from the two coarser fractions under a binocular microscope to avoid weathered surfaces and cloudy grains. Except for the garnet separate (CO-024), separated minerals were optically pure apart from variable amounts of spinel inclusions that could not be avoided. Their effect was tested for amphibole by preparing two distinct aliquots, one of coarse grains (>600 μm) with abundant inclusions and one finer-grained (100–600 μm) with fewer inclusions, for the same sample (CO-004-A). A subset of mineral separates was also analysed after leaching in 6 N HCl for 2 h at 80 °C, although the sample disaggregation seemed to have mainly produced grains with fresh surfaces. In addition, a thinly layered sample exhibiting progressive LREE enrichment (Tilhac et al., 2016) was cut into three sections parallel to the layering (Fig. 4a and b), digested and analysed following the same procedures as for whole-rock samples. Mineral separates and whole-rock samples were digested and processed using ion-exchange techniques described by Pin and Santos Zalduegui (1997), with one column for amphibole and garnet (~100–150 mg per sample) and 2–3 columns per sample for cpx

and whole rock (up to ~450 mg), depending on their Nd concentration. Analytical procedures are detailed in Electronic Appendix 1.

Sr and Nd isotopic ratios were obtained by thermal ionization mass spectrometry (TIMS) on a Thermo Finnigan Triton system. SRM-987 and JMC-321 were measured to check instrument status and sensitivity for Sr and Nd; they yielded $^{87}\text{Sr}/^{86}\text{Sr} = 0.710218 \pm 3$ and $^{143}\text{Nd}/^{144}\text{Nd} = 0.511128 \pm 2$ (in-run errors), respectively, which are within error of the GeoRem recommended values (<http://georem.mpch-mainz.gwdg.de>). Reference samples BHVO-2 and BCR-2, prepared in the same way as the samples, were analysed as unknown. Three analyses of BCR-2 yielded $^{87}\text{Sr}/^{86}\text{Sr} = 0.704998 \pm 6$ (2σ) and $^{143}\text{Nd}/^{144}\text{Nd} = 0.512634 \pm 7$ (2σ) and three analyses of BHVO-2 yielded $^{87}\text{Sr}/^{86}\text{Sr} = 0.703477 \pm 30$ (2σ) and $^{143}\text{Nd}/^{144}\text{Nd} = 0.512987 \pm 20$ (2σ); these are comparable to the values reported by Jweda et al. (2016) for BCR-2 ($^{87}\text{Sr}/^{86}\text{Sr} = 0.705000 \pm 11$; $^{143}\text{Nd}/^{144}\text{Nd} = 0.512637 \pm 13$) and to GeoRem preferred values for BHVO-2 ($^{87}\text{Sr}/^{86}\text{Sr} = 0.703469 \pm 34$; $^{143}\text{Nd}/^{144}\text{Nd} = 0.512980 \pm 12$). Ratios were normalized to $^{86}\text{Sr}/^{88}\text{Sr} = 0.1194$ and $^{146}\text{Nd}/^{144}\text{Nd} = 0.7219$ to correct for mass fractionation. Blanks were generally below 1000 and 80 pg for Sr and Nd, respectively. $^{87}\text{Rb}/^{86}\text{Sr}$ and $^{147}\text{Sm}/^{144}\text{Nd}$ were calculated from their relative isotopic abundances and the elemental ratios measured by laser ablation inductively-coupled plasma mass spectrometry (LA-ICP-MS) for cpx and amphibole, and by solution ICP-MS for whole rock and garnet (Tilhac et al., 2016).

3.3. REE thermometry and diffusion modelling

The timescale for diffusive re-equilibration (t_D) of the REE (HREE) was calculated from the equation given by Liang (2014), as detailed in Electronic Appendix 2a. Calculations were performed at the sub-solidus temperature given by the REE-in-two-pyroxene thermometer of Liang et al. (2013) restricted to the HREE and Y to avoid bias related to secondary processes preferentially affecting the LREE (Electronic Appendix 2b). In addition, to account for various degrees of amphibolitization and thus of sub-solidus re-equilibration of cpx with amphibole, the REE compositions of cpx used for REE thermometry were corrected by re-incorporating the REE content of amphibole into cpx (Electronic Appendix 2c). Modal compositions, major- and trace-element compositions of cpx, opx and amphibole and major-element-based thermobarometric calculations have been reported by Tilhac et al. (2016).

4. Results

4.1. Sr- and Nd-isotope geochemistry

Sr-isotope compositions are reported in Table 1. Cpx from all types of pyroxenite have very low $^{87}\text{Rb}/^{86}\text{Sr}$ (<0.015) and $^{87}\text{Sr}/^{86}\text{Sr}$ ranging between 0.7037 and 0.7045 (Fig. 3a), with most cpx in type-3 pyroxenites restricted to radiogenic values (~ 0.7045). Leached separates yield similar $^{87}\text{Sr}/^{86}\text{Sr}$ within $\pm 2 \times 10^{-5}$, or slightly more radiogenic Sr (by less than 3×10^{-4}) than their non-leached replicates. This confirms that most grains were free of surface contamination in terms of Sr isotopes, indicating the reliability of the data collected on non-leached minerals. One exception has more radiogenic Sr in the leached minerals, although the corresponding error is very high ($\pm 113 \times 10^{-6}$); probably due to the removal of surface contaminants with non-radiogenic Sr, such as carbonates. When compared with corresponding cpx, type-1 and -2 amphiboles have significantly more radiogenic values and higher $^{87}\text{Rb}/^{86}\text{Sr}$, whereas type-3 cpx and amphiboles are closer to equilibrium, with only slightly higher $^{87}\text{Rb}/^{86}\text{Sr}$ and $^{87}\text{Sr}/^{86}\text{Sr}$ (Fig. 3a). Leached and non-leached amphibole separates give very similar Sr-isotope compositions, as do the fine-grained and coarse-grained (with more abundant spinel inclusions) amphibole separates. This

indicates the reliability of the data collected on non-leached amphibole and confirms that spinel inclusions do not affect the Sr isotopes. Most whole rocks have higher $^{87}\text{Rb}/^{86}\text{Sr}$ and more radiogenic $^{87}\text{Sr}/^{86}\text{Sr}$ than cpx, and plot close to the mixing lines between cpx and amphibole (or cpx and garnet for CO-024), indicating that no minor phase is needed to account for the whole-rock data. Garnet separated from sample CO-024 is significantly more radiogenic than the corresponding cpx and whole rock. Among the three sections of the sample exhibiting progressive LREE enrichment (CO-010), the most LREE enriched ones (1 and 2) have relatively non-radiogenic Sr (0.7040–0.7041) compared to type-3 pyroxenites (Fig. 4c), while the least enriched one (3) exhibits a higher $^{87}\text{Sr}/^{86}\text{Sr}$ (0.7043). Sr isotopes in harzburgites are significantly more radiogenic than in any pyroxenites.

Nd-isotope compositions are reported in Table 2. Cpx, amphibole and whole-rock compositions from the different types of pyroxenite have distinctive Nd-isotope compositions (Fig. 3b). All type-3 pyroxenites plot within a narrow range of low $^{143}\text{Nd}/^{144}\text{Nd}$ (0.5125–0.5127) and low $^{147}\text{Sm}/^{144}\text{Nd}$ (0.08–0.16) while most type-1 pyroxenites have relatively radiogenic $^{143}\text{Nd}/^{144}\text{Nd}$ (0.5130–0.5133) at higher $^{147}\text{Sm}/^{144}\text{Nd}$ (0.15–0.31). Two exceptions are LREE-enriched type-1 samples, one of which was sampled near a pyroxenite dyke; these plot within the field of type-3 pyroxenites. Regarding type-2 pyroxenites, the least LREE-enriched samples plot between type-1 and -3 pyroxenites while the most enriched ones overlap with type-3 pyroxenites. The Nd-isotope composition of the LREE-enriched type-4 pyroxenite is similar to that of type-3 pyroxenites while the LREE-depleted one exhibits $^{147}\text{Sm}/^{144}\text{Nd}$ (0.33) similar to type-1 pyroxenites, and moderately radiogenic $^{143}\text{Nd}/^{144}\text{Nd}$. By contrast with Sr, Nd in amphibole is only slightly more radiogenic and $^{147}\text{Sm}/^{144}\text{Nd}$ higher than in the corresponding cpx, and this difference is even smaller in the case of type-3 pyroxenites. Leached cpx and amphibole separates are within error of the non-leached replicates or have slightly more radiogenic Nd (less than a 15×10^{-5} difference in $^{143}\text{Nd}/^{144}\text{Nd}$), indicating the reliability of the data collected on non-leached minerals. If anything, leaching has removed a surface contaminant with slightly less radiogenic Nd. Whole rocks have either more radiogenic or similar Nd than cpx and exhibits intermediate $^{147}\text{Sm}/^{144}\text{Nd}$ between cpx and amphibole, where data are available. For the sample investigated, garnet has expectedly more radiogenic Nd with higher $^{147}\text{Sm}/^{144}\text{Nd}$ than cpx (and whole rock). The Nd-isotope compositions of the three sections of sample CO-010 remain within the field of type-3 pyroxenites (Fig. 3b). As for Sr, section 1 and 2 have similar $^{143}\text{Nd}/^{144}\text{Nd}$ (~ 0.5125 ; Fig. 4d), which is accordingly less radiogenic than that of the least LREE-enriched section 3 (0.5127). The two analysed harzburgites correspond respectively to the most and least radiogenic samples of the whole dataset, with the highest and lowest $^{147}\text{Sm}/^{144}\text{Nd}$.

On a plot of present-day ε_{Nd} vs $^{87}\text{Sr}/^{86}\text{Sr}$ (Fig. 5a), most samples of cpx, amphibole and whole rock are comparable to data previously reported for Cabo Ortegal pyroxenites (Gravestock, 1992), which plot near or within the field of ocean island basalts (OIB; Hofmann, 2014), extending to higher and lower ε_{Nd} with type-1 and type-3 (and other LREE-enriched) pyroxenites, respectively. With respect to the field of OIB, these samples plot close to the enriched mantle (EM I) end member. Cabo Ortegal pyroxenites thus have heterogeneous present-day Nd-isotope compositions (across $\sim 20\varepsilon_{\text{Nd}}$ units) over a relatively narrow range of Sr-isotope compositions. Compared with pyroxenites, Cabo Ortegal eclogites have homogeneous Sr- and Nd-isotope compositions, with ε_{Nd} similar to those of type-1 pyroxenites over a range of $^{87}\text{Sr}/^{86}\text{Sr}$. However, like pyroxenites, Cabo Ortegal granulites and garnet-bearing mafic rocks occurring near the top of the pyroxenite-rich domain are more scattered in present-day Sr-Nd space (Fig. 5a).

Table 1
Sr-isotope data for Cabo Ortegal pyroxenites and peridotites.

Sample		Rb (ppm)	Sr (ppm)	$^{87}\text{Rb}/^{86}\text{Sr}$	$^{87}\text{Sr}/^{86}\text{Sr}$	$\pm$	$^{87}\text{Sr}/^{86}\text{Sr}$ at 390 Ma
<i>Type-1 pyroxenites</i>							
CO-091-A	cpx	0.02	51	0.0014	0.704245	2	0.704237
CO-094-B	cpx	0.02	9	0.0066	0.704411	3	0.704374
CO-095-A	cpx	0.04	34	0.0035	0.703946	3	0.703926
	leached cpx	0.04	34	0.0035	0.704134	51	0.704115
CO-096-B	cpx	0.02	22	0.0027	0.704276	2	0.704261
	leached cpx	0.02	22	0.0027	0.704310	5	0.704295
	amph	2.38	59	0.1169	0.704816	3	0.704167
	WR	0.46	16	0.0819	0.704586	3	0.704131
<i>Type-2 pyroxenites</i>							
CO-006-A	cpx	0.07	13	0.0153	0.704370	2	0.704285
	amph	1.74	28	0.1793	0.705144	3	0.704148
CO-006-B	cpx	0.09	92	0.0028	0.704301	3	0.704286
	WR	0.61	59	0.0300	0.704329	2	0.704162
CO-007	cpx	0.04	18	0.0059	0.703727	2	0.703694
	leached cpx	0.04	18	0.0059	0.704032	18	0.703999
CO-024	cpx	0.02	25	0.0019	0.703975	3	0.703964
	leached cpx	0.02	25	0.0019	0.703922	9	0.703911
	grt	0.25	8	0.0899	0.704259	3	0.704210
	WR	0.78	42	0.0540	0.704154	3	0.703854
CO-025	cpx	0.02	58	0.0012	0.703949	2	0.703942
	leached cpx	0.02	58	0.0012	0.703952	9	0.703945
	WR	0.65	90	0.0210	0.704045	2	0.703929
<i>Type-3 pyroxenites</i>							
CO-004-A	cpx	0.03	345	0.0003	0.704455	3	0.704453
	amph	4.43	558	0.0230	0.704551	2	0.704424
	amph*	4.43	558	0.0230	0.704557	3	0.704430
	WR	1.58	425	0.0108	0.704458	2	0.704398
CO-009	cpx	0.02	75	0.0007	0.703966	3	0.703962
	leached cpx	0.02	75	0.0007	0.705806	113	0.705801
	amph	2.31	210	0.0318	0.703998	4	0.703980
	leached amph	2.31	210	0.0318	0.703984	3	0.703808
CO-010	section 1	0.71	160	0.0128	0.704021	8	0.704014
	section 2	0.32	77	0.0120	0.704091	3	0.704084
	section 3	0.25	33	0.0217	0.704277	4	0.704265
CO-067	cpx	0.25	217	0.0034	0.704470	2	0.704451
CO-101	cpx	0.02	98	0.0007	0.704410	2	0.704406
<i>Type-4 pyroxenites</i>							
CO-002-A	cpx	0.02	19	0.0023	0.704128	3	0.704116
CO-048	cpx	0.03	261	0.0003	0.704244	3	0.704242
<i>Harzburgites</i>							
CO-015-B	WR	0.25	20	0.0365	0.705300	73	0.705097
CO-084	WR	0.29	3	0.2720	0.705739	3	0.704228

WR, whole rock; cpx, clinopyroxene; amph, amphibole. *coarse ($>600\text{ }\mu\text{m}$) amphibole separate. Errors are given for the last digit on $^{87}\text{Sr}/^{86}\text{Sr}$. An accepted decay constant ($\lambda_{\text{Rb}} = 142 \times 10^{-11} \text{ a}^{-1}$) was used in the calculation of age-corrected ratios at $t = 390\text{ Ma}$. See text and Electronic Appendix 1 for further details on sample preparation and analytical procedures.

Age-corrected ratios at $t = 390\text{ Ma}$ are more homogeneous, regardless of their type and of the nature of the sample (cpx, amphibole, garnet or whole rock); the comparison with present-day ratios on identical scales is striking (Fig. 5b). $\varepsilon_{\text{Nd}(t)}$ range between 0 and +7.5 and $^{87}\text{Sr}/^{86}\text{Sr}(t)$ between 0.7037 and 0.7045, with the exception of one harzburgite. These compositions lie between those of the depleted MORB mantle (DMM) and the enriched mantle components (EM I/II).

4.2. Sr and Nd geochronology

External (i.e. between samples) and internal (within individual samples) isochrons have been calculated for both the Rb–Sr and the Sm–Nd isotopic systems (Electronic Appendix 3a). On an $^{87}\text{Rb}/^{86}\text{Sr}$ vs $^{87}\text{Sr}/^{86}\text{Sr}$ plot (Fig. 3a), regardless of the type of pyroxenite but with a number of exceptions, the isotopic compositions of amphibole and whole rock, and to a lesser extent of cpx, plot on a line which includes one of the harzburgite samples as its most radiogenic end member. This line would give an isochron age of $463 \pm 65\text{ Ma}$. Internal isochrons between cpx and amphibole (or garnet) give ages between 229 and 332 Ma. On a

plot of $^{147}\text{Sm}/^{144}\text{Nd}$ vs $^{143}\text{Nd}/^{144}\text{Nd}$ (Fig. 3b), all types of pyroxenite fall on a line whose slope would equate an isochron age of $474 \pm 59\text{ Ma}$ with an initial $^{143}\text{Nd}/^{144}\text{Nd}$ of ~ 0.5122 ($\varepsilon_{\text{Nd}} = -7.8$). However, the scatter of the data is relatively significant for samples with high $^{147}\text{Sm}/^{144}\text{Nd}$, essentially due to amphiboles, which would give an isochron age of $332 \pm 170\text{ Ma}$, as discussed below. Cpx-amphibole pairs give isochron ages of 322 and 352 Ma, where data are available. Isochron ages have also been calculated for *type-2* pyroxenites using cpx-whole-rock (515 Ma) and cpx, garnet and whole rock (280 Ma). In addition, the increasingly radiogenic and LREE-enriched sections of sample CO-010 give a relatively well-defined Nd isochron age of $459 \pm 84\text{ Ma}$ (Fig. 4d), while Rb–Sr gives unrealistic errorchron ages ($>1.2\text{ Ga}$; Fig. 4c).

Depleted-mantle model ages (τ_{DM}) calculated from Nd-isotope compositions give two different groups (Table 2). The first one corresponds to the cluster of radiogenic samples (LREE-depleted *type-1* and *-2* pyroxenites) and gives mostly meaningless future τ_{DM} , while the other group gives a range of very high ages (669–1583 Ma), corresponding to the cluster of non-radiogenic, LREE-enriched samples. Nonetheless, considering the ingrowth of radiogenic Nd, as discussed below, these single-stage model ages

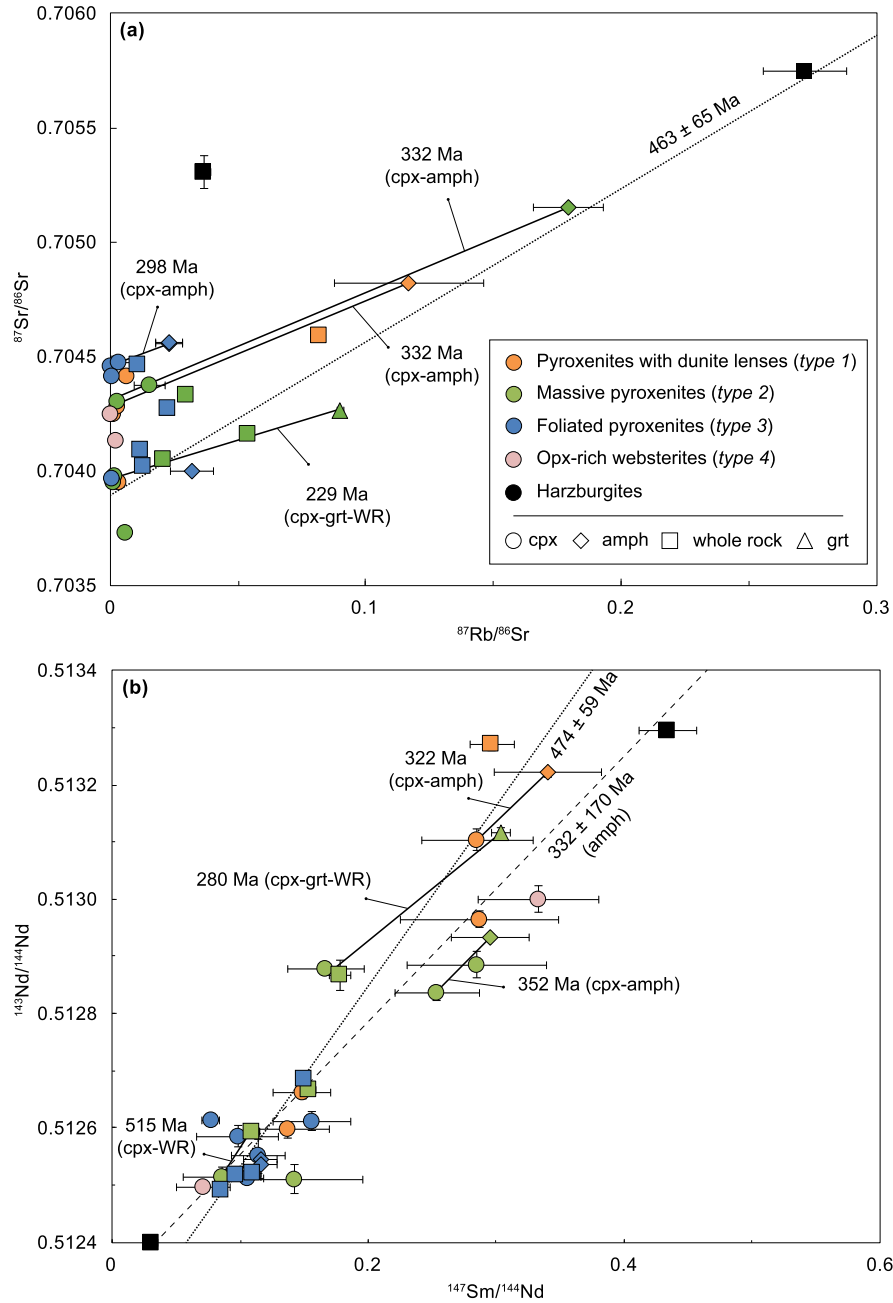


Fig. 3. Measured (a) $^{87}\text{Sr}/^{86}\text{Sr}$ vs $^{87}\text{Rb}/^{86}\text{Sr}$ and (b) $^{143}\text{Nd}/^{144}\text{Nd}$ vs $^{147}\text{Sm}/^{144}\text{Nd}$ for cpx, amphibole and whole rock of Cabo Ortegal pyroxenites (and peridotites), and possible internal (full lines) and external (dashed lines) Rb–Sr and Sm–Nd isochrons (see text and Electronic Appendix 3a for further details). Error used are in-run errors for $^{87}\text{Sr}/^{86}\text{Sr}$ and $^{143}\text{Nd}/^{144}\text{Nd}$ and standard deviations (2σ) for $^{87}\text{Rb}/^{86}\text{Sr}$ and $^{147}\text{Sm}/^{144}\text{Nd}$.

(i.e. calculated from present-day $^{147}\text{Sm}/^{144}\text{Nd}$ and $^{143}\text{Nd}/^{144}\text{Nd}$) are certainly meaningless. Model ages have thus been corrected for this radiogenic ingrowth by first calculating age-corrected $^{143}\text{Nd}/^{144}\text{Nd}$ at 390 Ma using present-day $^{147}\text{Sm}/^{144}\text{Nd}$, then minimum depleted-mantle model ages ($\tau_{\text{DM}(2)}$) from the age-corrected $^{143}\text{Nd}/^{144}\text{Nd}$, assuming $^{147}\text{Sm}/^{144}\text{Nd} = 0$. These $\tau_{\text{DM}(2)}$ are strikingly homogeneous when compared to the range of Nd compositions (Fig. 6a), yielding an average of 654 ± 72 Ma (1σ ; Table 2).

4.3. REE diffusional timescale

Temperatures provided by REE thermometry have been used to model the timescale of diffusional re-equilibration of the REE between cpx and opx. Temperature inversion diagrams for individ-

ual samples show that LREE deviate from the linear trend passing through the origin as a result of secondary processes (Liang et al., 2013), but show the overall robustness of the calculation based on the HREE and Y (Electronic Appendix 2b). This provides an average corrected T_{HREE} of 843 ± 55 °C (Fig. 6b), which is in relatively good agreement with major-element thermometric calculations, although these expectedly give lower values depending on the method used (Tilhac et al., 2016).

As defined by Liang (2014), the diffusional timescale t_D corresponds to the time required for the mineral with the slowest diffusivities (opx, in the case of HREE) to eliminate significant compositional gradient within grains. As illustrated for Yb, t_D expectedly shows a strong dependency on the sub-solidus temperature (Fig. 6b); the higher the temperature, the more rapidly diffusive re-equilibration occurs. Thus, in order to provide a minimum esti-

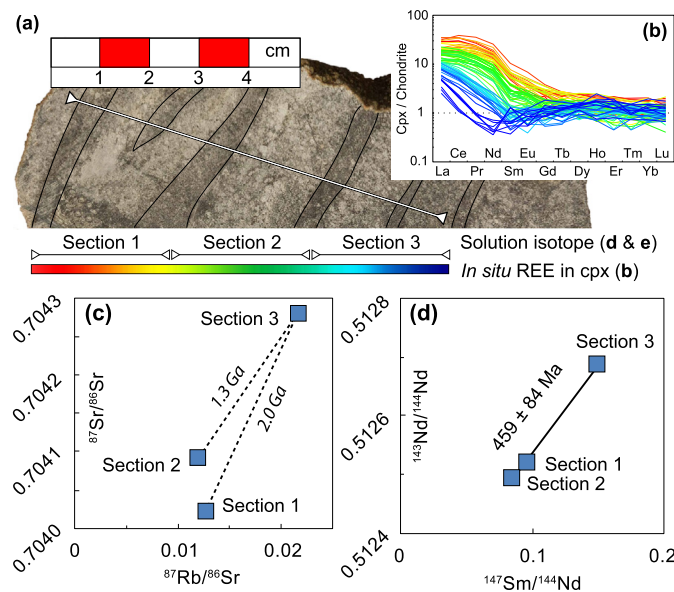


Fig. 4. Details of the REE and Sr- and Nd-isotope analysis of an increasingly LREE-enriched sample (CO-010), showing the location (a) of the LA-ICP-MS profile of REE compositions in cpx and (b) of the three sections analyzed for (c) Sr- and (d) Nd-isotope compositions with corresponding isochrons. Note that the unrealistic Rb–Sr ages are probably due to the sensitivity of low-Sr (i.e. high $^{87}\text{Rb}/^{86}\text{Sr}$) samples to late addition of radiogenic Sr. See text and Tilhac et al. (2016) for further details.

mate of this diffusional timescale, we assumed the maximum sub-solidus temperature provided within 1σ error by REE thermometry (899 °C). For grain sizes of 1.0–1.5 mm, representative of porphyroclasts (and coarse granoblasts) of Cabo Ortegal pyroxenites (Tilhac et al., 2016; Henry et al., 2017) on which the REE contents were analysed, t_D range between 403 and 906 Ma (Fig. 6b). The reliability and significance of these values are discussed below.

5. Discussion

5.1. Isotopic constraints on petrological processes

The positive correlation between LREE enrichment and Nd-isotope compositions allows for a relatively good discrimination between the different types of pyroxenite exposed in the Cabo Ortegal Complex (Fig. 3b). This implies that their range of $^{143}\text{Nd}/^{144}\text{Nd}$ is essentially related to the ingrowth of Nd over time, which is consistent with the relatively homogeneous age-corrected $^{143}\text{Nd}/^{144}\text{Nd}$ ($^{143}\text{Nd}/^{144}\text{Nd}_t$ at $t = 390$ Ma). The value of 390 Ma previously used for the age correction of Cabo Ortegal pyroxenites (Santos et al., 2002) is supported by the inversion of our data, which shows that an age of 377 Ma yields the minimum dispersion of $^{143}\text{Nd}/^{144}\text{Nd}_t$ (Electronic Appendix 3b).

A striking feature of the Nd-isotope compositions of Cabo Ortegal pyroxenites is the contrast between the homogeneity of low- $^{147}\text{Sm}/^{144}\text{Nd}$ data and the dispersion of data with high $^{147}\text{Sm}/^{144}\text{Nd}$ (Fig. 3b). This contrast can be satisfactorily accounted for by a two-stage process in which isotopic equilibrium was initially reached (e.g. after the pyroxenites formation), then followed by isotopic perturbations and/or changes in $^{147}\text{Sm}/^{144}\text{Nd}$. Assuming such a process, the initial isotopic equilibrium could be ascribed, in the light of the petrogenetic model proposed by Tilhac et al. (2016), to a late-magmatic episode of re-equilibration with trapped or percolating residual melts. This is in good agreement with the consistency between the internal isochrons calculated for a cpx-whole-rock pair and the different sections of a progressively LREE-enriched sample, and the external isochron calculated for all types of pyroxenite (459–515 Ma). Isotopic perturbations, and notably the homogenization of the LREE-enriched samples (especially type-3 pyroxenites), may have

occurred during the metamorphic amphibolitization of Cabo Ortegal pyroxenites, which is evidenced by field, petrographic and geochemical observations (Tilhac et al., 2016; Henry et al., 2017; Tilhac, 2017). This was potentially accompanied by slight changes in the partitioning of Sm and Nd between cpx and amphibole, either due to the flattening of the LREE in amphibole reaching equilibrium with percolating fluids, and/or to additional LREE enrichment in cpx (Electronic Appendix 4). Both of these alternatives are consistent with a metamorphic amphibolitization and may relate to the percolation of hydrous fluids, relatively unradiogenic and LREE-enriched. This is also in good agreement with the consistency between internal Nd isochrons calculated for cpx-amphibole pairs and the external isochron calculated using amphiboles of all types of pyroxenite (322–352 Ma), which confirms that amphibole and cpx were initially in isotopic equilibrium. Consequently, the present-day difference in $^{143}\text{Nd}/^{144}\text{Nd}$ between cpx and amphibole may result only from radiogenic ingrowth at slightly different $^{147}\text{Sm}/^{144}\text{Nd}$.

Sr-isotope systematics, unlike Nd, do not show a straightforward correlation between isotopic and parent–daughter ratios, because the strong partitioning of Rb into amphibole has resulted in very low $^{87}\text{Rb}/^{86}\text{Sr}$ in cpx and thus higher $^{87}\text{Rb}/^{86}\text{Sr}$ in amphibole (Electronic Appendix 4). However, this is still consistent with the two-stage scenario outlined from the Nd-isotope systematics. Most amphiboles and whole rocks indeed plot along a line which corresponds to an isochron age of 463 ± 65 Ma, while cpx either plot on this line or deviate from it towards higher $^{87}\text{Sr}/^{86}\text{Sr}$, as illustrated by samples of type-3 pyroxenites (Fig. 3a). Consequently, an initial isotopic equilibrium between cpx and amphibole may be envisaged provided that a radiogenic component was subsequently added to cpx. While the initial equilibrium may correspond to the late-magmatic metasomatic re-equilibration of Cabo Ortegal pyroxenites, the addition of a radiogenic component probably relates to their metamorphic amphibolitization. This is consistent with the range of internal isochron ages (298–323 Ma) calculated for cpx-amphibole pairs. No Rb–Sr isochron could be calculated between the different amphiboles due to the significant radiogenic ingrowth of Sr that occurred between the late-magmatic/metasomatic episode and the metamorphic amphibolitization.

Sr- and Nd-isotope systematics consistently indicate two main episodes during which cpx and amphibole have acquired their

Table 2

Nd-isotope data for Cabo Ortegal pyroxenites and peridotites.

Sample		Sm (ppm)	Nd (ppm)	$^{147}\text{Sm}/^{144}\text{Nd}$	$^{143}\text{Nd}/^{144}\text{Nd}$	$\pm$	ε_{Nd}	τ_{DM} (Ma)	$^{143}\text{Nd}/^{144}\text{Nd}$ at 390 Ma	ε_{Nd} at 390 Ma	$\tau_{\text{DM}(2)}$ (Ma)
<i>Type-1 pyroxenites</i>											
C0-091-A	cpx	0.30	1.24	0.148	0.512662	10	0.6	1285	0.512284	3.0	665
C0-094-B	cpx	0.28	0.58	0.287	0.512965	15	6.5	−523	0.512232	2.0	702
C0-095-A	cpx	0.21	0.92	0.137	0.512598	15	−0.6	1231	0.512247	2.3	691
	leached cpx	0.21	0.92	0.137	0.512595	8	−0.7	1236	0.512245	2.2	693
C0-096-B	cpx	0.13	0.28	0.285	0.513103	19	9.2	−238	0.512374	4.8	600
	leached cpx	0.13	0.28	0.285	0.513251	10	12.1	76	0.512522	7.7	495
	amph	0.68	1.21	0.341	0.513221	8	11.5	7	0.512350	4.3	618
	WR	0.12	0.24	0.297	0.513271	12	12.5	102	0.512512	7.5	502
<i>Type-2 pyroxenites</i>											
C0-006-A	cpx	0.12	0.29	0.254	0.512837	13	4.0	−1453	0.512189	1.2	733
	amph	0.55	1.13	0.296	0.512933	4	5.9	−527	0.512178	1.0	740
C0-006-B	cpx	0.49	2.07	0.143	0.512511	25	−2.3	1508	0.512147	0.3	762
	WR	0.51	2.00	0.154	0.512666	5	0.7	1389	0.512274	2.8	672
C0-007	cpx	0.11	0.22	0.285	0.512886	24	5.0	−708	0.512158	0.6	754
	leached cpx	0.11	0.22	0.285	0.512891	13	5.1	−697	0.512163	0.7	751
C0-024	cpx	0.26	0.93	0.167	0.512878	10	4.8	1094	0.512452	6.3	545
	leached cpx	0.26	0.93	0.167	0.512937	3	6.0	903	0.512511	7.4	503
	grt	0.10	0.19	0.304	0.513116	10	9.5	−169	0.512339	4.1	625
	WR	0.30	1.01	0.178	0.512866	26	4.6	1500	0.512411	5.5	574
C0-025	cpx	0.38	2.72	0.086	0.512515	17	−2.2	833	0.512297	3.3	656
	leached cpx	0.38	2.72	0.086	0.512609	3	−0.4	722	0.512390	5.1	589
	WR	0.89	4.91	0.109	0.512595	2	−0.7	905	0.512316	3.6	642
<i>Type-3 pyroxenites</i>											
C0-004-A	cpx	0.48	2.74	0.106	0.512512	4	−2.3	990	0.512243	2.2	694
	amph	2.07	10.80	0.116	0.512545	2	−1.7	1042	0.512250	2.3	689
	amph*	2.07	10.80	0.116	0.512537	3	−1.8	1054	0.512242	2.2	695
	WR	1.22	6.68	0.110	0.512523	5	−2.1	1021	0.512241	2.2	695
C0-009	cpx	0.19	1.17	0.098	0.512586	19	−0.9	828	0.512336	4.0	627
	leached cpx	0.19	1.17	0.098	0.512580	18	−1.0	836	0.512330	3.9	632
	amph	2.81	22.11	0.077	0.512614	5	−0.3	669	0.512418	5.6	569
	leached amph	2.81	22.11	0.077	0.512605	6	−0.5	680	0.512409	5.5	576
C0-010	section 1	1.41	8.86	0.096	0.512520	2	−2.1	903	0.512274	2.8	672
	section 2	0.36	2.58	0.085	0.512494	2	−2.7	853	0.512277	2.9	669
	section 3	0.10	0.42	0.149	0.512687	12	1.1	1246	0.512306	3.5	649
C0-067	cpx	0.24	1.30	0.114	0.512552	29	−1.5	1014	0.512260	2.6	681
C0-101	cpx	0.44	1.73	0.156	0.512613	17	−0.3	1583	0.512215	1.7	714
<i>Type-4 pyroxenites</i>											
C0-002-A	cpx	0.08	0.15	0.334	0.513000	24	7.2	−275	0.512148	0	761
C0-048	cpx	0.50	4.23	0.071	0.512497	1	−2.6	767	0.512316	3.6	642
<i>Harzburgites</i>											
C0-015-B	WR	0.01	0.24	0.030	0.512401	8	−4.5	675	0.512325	3.8	636
C0-084	WR	0.05	0.07	0.435	0.513294	10	13.0	55	0.512184	1.1	736

WR, whole rock; cpx, clinopyroxene; amph, amphibole. *coarse (>600 μm) amphibole separate. τ_{DM} and $\tau_{\text{DM}(2)}$, single- and second-stage depleted mantle model ages, respectively. Errors are given for the last digit on $^{143}\text{Nd}/^{144}\text{Nd}$. An accepted decay constant ($\lambda_{\text{Sm}} = 6.54 \times 10^{-12} \text{ a}^{-1}$) was used in the calculation of age-corrected ratios at $t = 390 \text{ Ma}$. ε_{Nd} were calculated as $[(^{143}\text{Nd}/^{144}\text{Nd}_{\text{sample}})/(^{143}\text{Nd}/^{144}\text{Nd}_{\text{CHUR}}) - 1] \times 10000$ with present-day CHUR ratios taken as $^{147}\text{Sm}/^{144}\text{Nd} = 0.1960$ and $^{143}\text{Nd}/^{144}\text{Nd} = 0.512630$ (Bouvier et al., 2008). τ_{DM} and $\tau_{\text{DM}(2)}$ were calculated using present-day depleted-mantle values of $^{147}\text{Sm}/^{144}\text{Nd} = 0.2137$ and $^{143}\text{Nd}/^{144}\text{Nd} = 0.513215$ (Hofmann, 2014; and references therein). See text and Electronic Appendix 1 for further details on sample preparation, analytical procedures and calculations.

disequilibrium, producing the range of isotope compositions now recorded in Cabo Ortegal pyroxenites. This is supported by the overall agreement between the compositions of leached and unleached minerals, implying that exotic Sr and Nd were introduced into the minerals structure through recrystallization and/or diffusion, and are probably not surficial. Sr and Nd isotopes are thus in good agreement with petrological and geochemical constraints (Gravestock, 1992; Tilhac et al., 2016; Henry et al., 2017; Tilhac, 2017), supporting the percolative fractional-crystallization model in which amphibole crystallized following the late-magmatic migration of residual melts, in isotopic equilibrium with the LREE-enriched pyroxenites.

5.2. Source constraints

Sr- and Nd-isotope compositions reported here are consistent with previously published values (Peucat et al., 1990; Gravestock, 1992; Santos et al., 2002). The overlap in Sr–Nd space between Cabo Ortegal pyroxenites, granulites and garnet-bearing mafic rocks from the same complex suggests that they are likely to be

cogenetic and/or to have crystallized from a parental melt that sampled the same source region. However, these mafic lithologies differ isotopically from Cabo Ortegal eclogites, which have a homogeneously depleted signature, probably inherited from their MORB-like basaltic/gabbroic protoliths (Fig. 5a). Correcting for radiogenic ingrowth over 390 Ma provides relatively homogeneous age-corrected $^{143}\text{Nd}/^{144}\text{Nd}$ and $^{87}\text{Sr}/^{86}\text{Sr}$ (Fig. 5b), mostly ranging between 0.5121 ($\varepsilon_{\text{Nd}} = 0$) and 0.5125 ($\varepsilon_{\text{Nd}} = +7.5$) and between 0.7037 and 0.7044, respectively. In other words, the present-day $^{147}\text{Sm}/^{144}\text{Nd}$ is sufficient to account for the range of $^{143}\text{Nd}/^{144}\text{Nd}$. The reliability of these age-corrected ratios (and thus of their homogeneity) is supported by their close similarity with the initial ratios of the different isochrons (Electronic Appendix 3a). The reliability of age-corrected Sr-isotope compositions is strengthened by the strong partitioning of Rb between cpx and amphibole rendering negligible the radiogenic ingrowth of Sr in cpx. A first-order implication of the homogeneity of the age-corrected compositions of Cabo Ortegal pyroxenites is that it supports the interpretation of the different types of pyroxenite as cogenetic (Tilhac et al., 2016). This also suggests that the pyroxenites do not represent the meta-

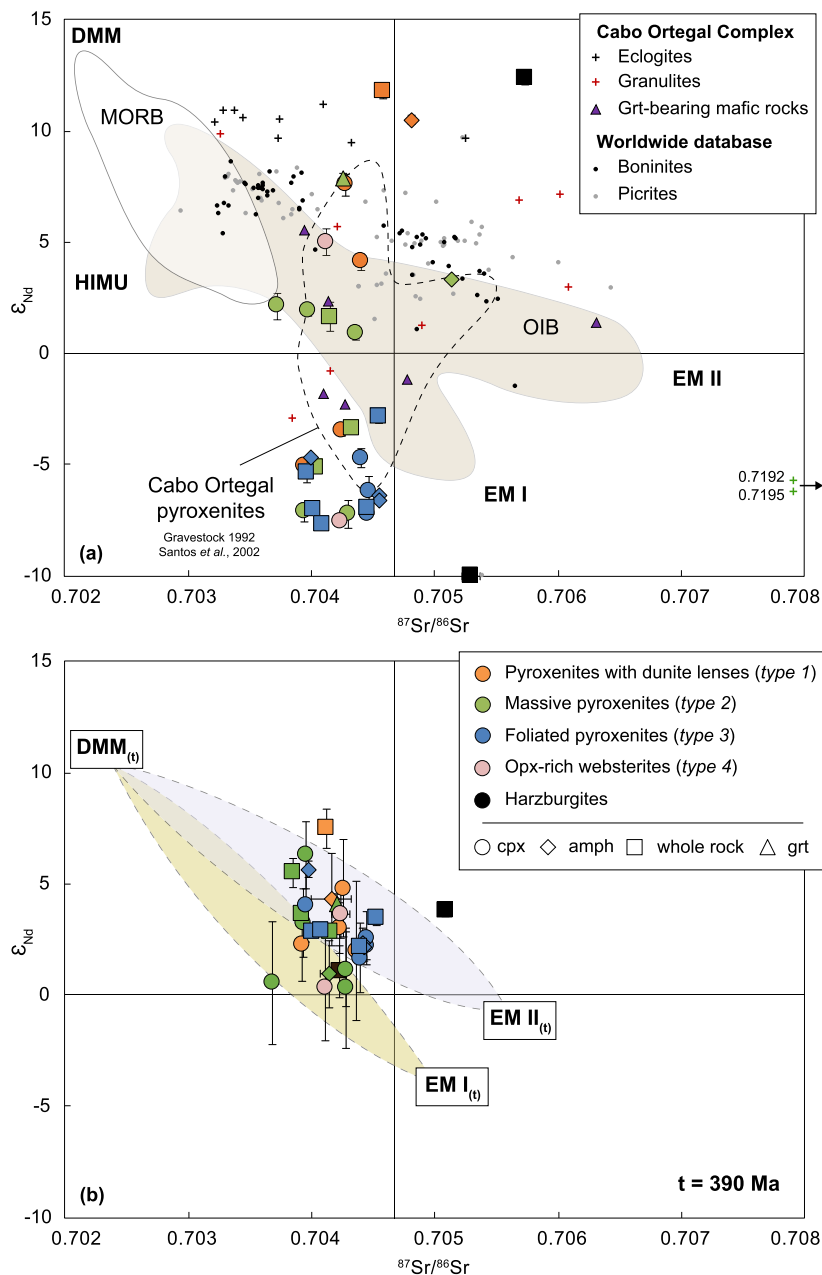


Fig. 5. (a) Present-day and (b) age-corrected ϵ_{Nd} vs $^{87}\text{Sr}/^{86}\text{Sr}$ for non-leached cpx, amphibole and whole-rock samples of Cabo Ortegal pyroxenites (and peridotites). Previous data on Cabo Ortegal pyroxenites (Gravestock, 1992; Santos et al., 2002), eclogites and granulites (Gravestock, 1992) and garnet-bearing mafic rocks (Peucat et al., 1990) are shown for comparison, along with a worldwide database of boninites and picrites sampled at convergent margins (Georock) and the compositional fields of MORB and OIB, with DMM, EM I, EM II and HIMU end members. Initial compositions of the oceanic reservoirs were calculated at 390 Ma from the present-day compositions of end-member lavas assuming: for DMM, $^{87}\text{Sr}/^{86}\text{Sr} = 0.7025$, $^{143}\text{Nd}/^{144}\text{Nd} = 0.513215$, $^{87}\text{Rb}/^{86}\text{Sr} = 0.022$ and $^{147}\text{Sm}/^{144}\text{Nd} = 0.2137$; for EM I, $^{87}\text{Sr}/^{86}\text{Sr} = 0.7052$, $^{143}\text{Nd}/^{144}\text{Nd} = 0.51240$, $^{87}\text{Rb}/^{86}\text{Sr} = 0.037$ and $^{147}\text{Sm}/^{144}\text{Nd} = 0.21$; for EM II, $^{87}\text{Sr}/^{86}\text{Sr} = 0.7074$, $^{143}\text{Nd}/^{144}\text{Nd} = 0.51255$, $^{87}\text{Rb}/^{86}\text{Sr} = 0.041$ and $^{147}\text{Sm}/^{144}\text{Nd} = 0.25$ (Hofmann, 2014; and references therein). The composition of CHUR was taken from Bouvier et al. (2008). Other symbols as in Fig. 3.

morphic products of low-pressure cumulates, which is consistent with the absence of a strongly positive Eu anomaly (e.g. Becker, 1996).

In the Cabo Ortegal Complex, relatively high age-corrected Pb-isotope ratios (compared to MORB) have been previously related to an island-arc origin with a continental component (Santos et al., 2002), while similarities of Sr and Nd compositions with arc volcanics were pointed out. In Sr–Nd space corrected to 390 Ma, our data on Cabo Ortegal pyroxenites are located between DMM and the enriched mantle reservoirs (EM I and EM II). The pyroxenite compositions may thus be accounted for by mixing these reservoirs. Assuming $([\text{Sr}]/[\text{Nd}])_{\text{EM}}/([\text{Sr}]/[\text{Nd}])_{\text{DMM}}$ ranging between 0.5 and 1.5, the respective mixing arrays between DMM and

EM I and II satisfactorily account for the whole range of age-corrected isotopic compositions (Fig. 5b). Contributions of EM I to the DMM have been variously interpreted elsewhere as the signature of recycled mantle-wedge peridotite, lower continental crust, pelagic sediments or delaminated subcontinental lithospheric mantle, while EM II contributions are mainly attributed to recycled sediments or a mixture of igneous oceanic crust and sediments (Kimura et al., 2016; and references therein). However, it has been increasingly accepted that the existence of different mantle components reflects only variations in the nature and scale of heterogeneity within specific domains rather than physically distinct reservoirs (Armienti and Gasperini, 2007; and references therein). Relatively radiogenic $^{87}\text{Sr}/^{86}\text{Sr}$ and negative ϵ_{Nd}

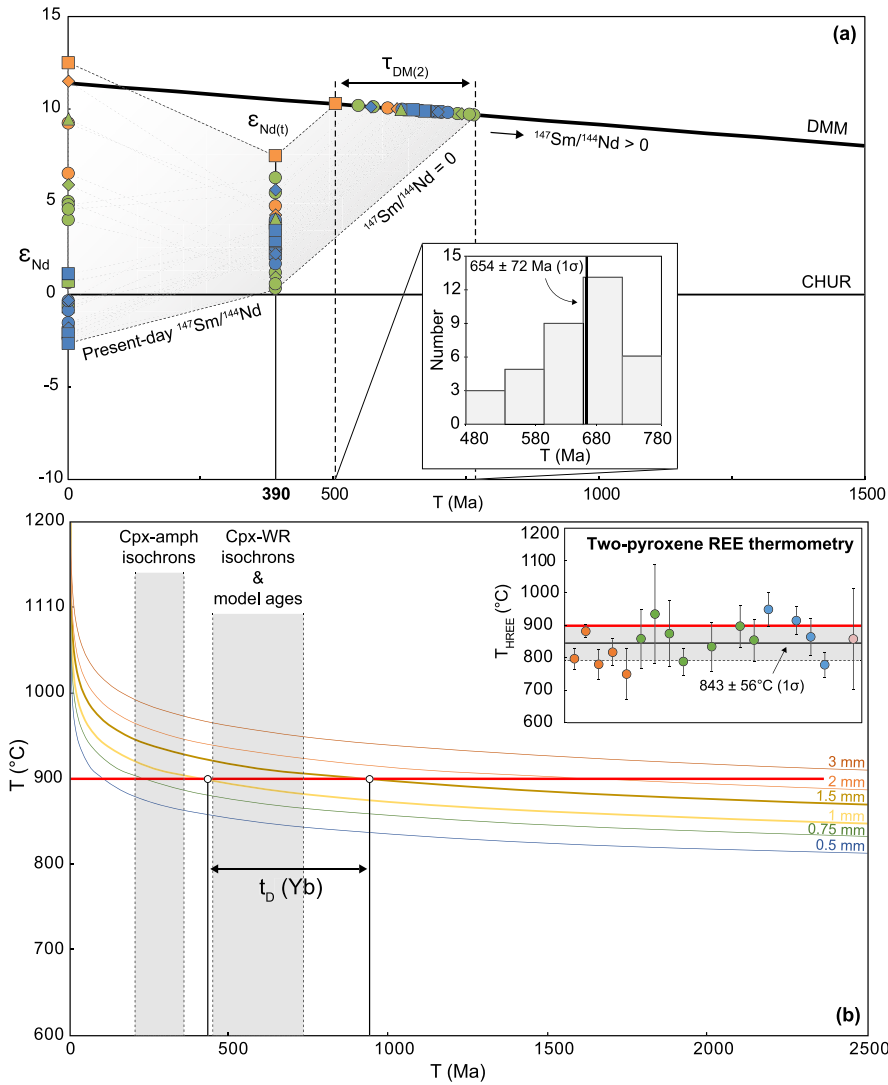


Fig. 6. (a) Time evolution of the Nd-isotope compositions of Cabo Ortegal pyroxenites showing calculated age-corrected ϵ_{Nd} and second-stage depleted-mantle model ages ($\tau_{DM(2)}$) with the corresponding histogram showing their distribution (insert). $\tau_{DM(2)}$ were calculated assuming the following present-day DM composition: $^{147}\text{Sm}/^{144}\text{Nd} = 0.2137$ and $^{143}\text{Nd}/^{144}\text{Nd} = 0.513215$ (see text for further details). Symbols as in Fig. 3. (b) Evolution of the diffusional timescale of Yb with temperature, for cpx and opx grain size ranging between 0.5 mm and 3.0 mm. The red line corresponds to the maximum two-pyroxene REE temperature measured within 1σ error (insert), for which t_D have been calculated assuming grain sizes of 1.0–1.5 mm for Cabo Ortegal pyroxenites, as detailed in Electronic Appendix 2a. Rb–Sr and Sm–Nd geochronological data are shown for comparison. Individual error bars correspond to internal standard deviation (1σ) for each temperature inversion (Electronic Appendix 2b). The different types of pyroxenite are distinguished as in Fig. 3.

such as those measured in the Cabo Ortegal pyroxenites have indeed been interpreted in other mantle pyroxenites as evidence for the involvement of a crustal component in their parental melts (e.g. Svojtka et al., 2016). This may be consistent with the positive correlation observed between Sr/Nd and Eu/Eu* in type-2 pyroxenites (Electronic Appendix 4), which can be accounted for by the incorporation of oceanic sediments. However, no correlation is observed between Eu/Eu* and $^{87}\text{Sr}/^{86}\text{Sr}$, probably invalidating this alternative. This slightly positive Eu anomaly may instead indicate the potential accumulation of plagioclase, which may have occurred either during low-pressure fractional crystallization, or the contribution of melts generated from a former plagioclase-bearing source (e.g. Becker, 1996). We favour the second alternative considering the similar normative compositions of all types of pyroxenite, the absence of a positive correlation between Eu/Eu* and the concentration of major incompatible elements (e.g. TiO_2 , Na_2O), and the low-pressure partial melting history inferred from the genesis of high $\text{CaO}/\text{Al}_2\text{O}_3$ melt (Tilhac et al., 2016).

5.3. Significance and implications of the geochronological data

The earliest tectono-metamorphic record in the Variscan belt of Europe is represented by eclogite- to granulites-facies metamorphism preserved in amphibolite-facies rocks, and associated with allochthonous mafic and ultramafic rocks and ophiolitic remnants (Matte, 1986). This HP–HT event is dated at 425–390 Ma in north-western Iberia (Martínez Catalán et al., 2009; and references therein), although it is diachronously recorded between the different structural units. In the Cabo Ortegal Complex, this corresponds to a relatively well-constrained magmatic event dated at ~ 390 Ma on cpx-garnet pairs from garnet pyroxenite dykes (Peucat et al., 1990; Santos Zalduegui et al., 1996, 2002), where sufficient amounts of garnet could be hand-picked to provide reliable internal isochrons. The HP–HT event was rapidly followed by retrograde metamorphism, notably under amphibolite-facies conditions, dated at 390–380 Ma in north-western Iberia (Dallmeyer et al., 1997). This event may correspond to the onset of the Gondwanan–Laurussian convergence following pre-Variscan oceanic subduction (Martínez Catalán et al., 2009).

Relating distinct episodes of partial melting and deformation to specific tectono-thermal events and determining the age and provenance of the different protoliths are problematic when studying the European Variscides (Kroner and Romer, 2013), especially for the exotic terranes that experienced HP–HT metamorphism (e.g. Peucat et al., 1990; Fernández-Suárez et al., 2007). In north-western Iberia, these terranes may likely constitute fragments of an ensialic island arc, potentially of Cambro-Ordovician age (Andonaegui et al., 2002; Santos et al., 2002), or older, as discussed below. A cluster of ages around 500 Ma has notably been reported in Cabo Ortegal (Van Calsteren et al., 1979; Santos et al., 2002), but it is poorly constrained, especially considering that some of the corresponding isochrons were calculated including harzburgites, pyroxenites and garnet-pyroxenite dykes, which certainly are of different ages. The significance of these isochrons thus rests only on the fact that they provide ages that are comparable to that of the protoliths of the associated metabasic rocks (Peucat et al., 1990; Ordóñez Casado et al., 2001). In other words, there are very limited time constraints on the formation of the main pyroxenites in the Cabo Ortegal Complex (i.e. those occurring parallel to the tectonic foliation), and therefore on the activity of the related magmatic arc. Our geochronological interpretations are therefore focused on this main set of pyroxenites.

The consistency between the Rb–Sr and Sm–Nd isotopic systems and that between internal and external isochrons indicate, to a certain extent, the reliability of the ages calculated in our study; two main clusters are observed. A cluster of old ages, whose most reliable values range between 459 and 515 Ma, is essentially defined by external isochrons (and a cpx-whole-rock pair). It includes an external Nd isochron calculated for all types of pyroxenite (474 ± 59 Ma) which remarkably coincides with an isochron calculated for three different sections of an increasingly LREE-enriched sample (459 ± 84) whose enrichment relates to late-magmatic metasomatism (Tilhac et al., 2016). The reliability of these ages is however weakened by the partial reset associated with amphibolitization, as discussed above, especially considering the greater sensitivity of low-Nd (i.e. high $^{147}\text{Sm}/^{144}\text{Nd}$) samples to isotopic perturbation, and thus suggesting that they should be considered as minimum ages. A similar comment can be made regarding the external Sr isochron calculated for a subset of all types of pyroxenite (463 ± 65 Ma). In addition, a cluster of young ages (229–352 Ma) is provided by amphiboles and cpx-amphibole pairs (Fig. 3). However, amphibolitization was probably accompanied by the addition of an enriched component as illustrated by type-3 pyroxenites exhibiting slightly more radiogenic Sr than samples with comparable $^{87}\text{Rb}/^{86}\text{Sr}$. As a consequence, these young ages may not be geologically (or quantitatively) meaningful. Similarly, the isochrons calculated from the garnet-bearing type-2 pyroxenite (CO-024) are young (229 and 280 Ma) compared to the expected metamorphic age corresponding to the peak conditions under which garnet crystallized (~ 390 Ma). Although the good correlation between cpx, garnet and whole rock seems to indicate the reliability of these ages, we believe that these are underestimated as the HP metamorphic episode was probably accompanied by the percolation of hydrous fluids (Tilhac et al., 2016).

In addition to isochron ages, we have calculated second-stage model ages $\tau_{\text{DM}(2)}$, assuming the absence of radiogenic ingrowth before 390 Ma (i.e. $^{147}\text{Sm}/^{144}\text{Nd} = 0$). This aims to approximate the minimum age of the separation between the parental melts of Cabo Ortegal pyroxenites and their assumed DMM-like source, as any radiogenic ingrowth would have yielded higher $\tau_{\text{DM}(2)}$. An alternative would have been to estimate the $^{147}\text{Sm}/^{144}\text{Nd}$ of the melt in equilibrium with LREE-enriched pyroxenites but this would have introduced an additional (and poorly constrained) assumption. $\tau_{\text{DM}(2)}$ calculated here (502–762 Ma; Fig. 6a) overlap with the cluster of old ages (459–515 Ma) obtained from the Sr and

Nd isochrons that are not related to amphibolitization, which supports the reliability of these ages and suggest that Cabo Ortegal pyroxenites formed in Cambro-Ordovician to Neoproterozoic times. In the light of regional geodynamics and petrological models, we propose that they record the reworking of a rifted continental margin during the building of a volcanic arc, potentially Cadomian, as previously suggested in north-western Iberia (e.g. Martínez Catalán et al., 2009; Andonaegui et al., 2016). A corollary of this interpretation is that previously reported ages around 390 Ma certainly correspond to a metamorphic event reflected by the amphibolitization of the main pyroxenites, although it may still be coeval with the intrusion of garnet-rich pyroxenite dykes (Santos et al., 2002). These ages are also mostly older than the previously poorly constrained event *ca* 500 Ma (Van Calsteren et al., 1979; Santos et al., 2002), suggesting that it could in turn represent the metamorphic record of an arc-continent collision during the formation of Gondwana, for instance (e.g. Fernández-Suárez et al., 2007).

5.4. On the age constraints of (arc) pyroxenites in orogenic/ophiolitic massifs

The diffusional timescale (t_D) calculated for Yb between 1.0 to 1.5 mm-sized pyroxenes that record a sub-solidus temperature of $\sim 900^\circ\text{C}$ overlap with the cluster of old ages obtained from Sr and Nd isochrons, and second-stage Nd model ages (Fig. 6b), and extend it to higher values (403–906 Ma). This may qualitatively support the reliability of our geochronological data, but mainly illustrates the fact that diffusional timescale calculations are, as expected, highly sensitive to the determination of a representative grain size and sub-solidus temperature. The error related to the determination of the commonly low concentration of REE in opx is especially critical as it propagates to the calculation of T_{THREE} . Diffusional timescale modelling of REE, such as reported by Le Roux et al. (2016), is thus unable to specify the timing of geological events within a few tens (or even hundreds) of millions of years, unless grain size and temperature can be determined within a few tenths of microns and degrees, respectively. Such requirements can be hardly met when studying deformed and/or metasomatized pyroxenites.

Considering that (1) diffusional timescale calculations only allow discrimination of events with a poor time resolution and relate to the sub-solidus re-equilibration of mineral assemblages, (2) Sr and Nd isochrons (and single-stage Nd model ages) are sensitive to late perturbations of LILE and LREE and, (3) both are very likely to occur during subduction-related and orogenic metamorphism, metasomatism and magmatic activity (e.g. France et al., 2015), we conclude that the use of second-stage model ages ($\tau_{\text{DM}(2)}$), and furthermore if consistent with isochron ages, is a robust way to envisage the timing of pyroxenite formation in the sub-arc mantle, and in the case of pyroxenites hosted in HP peridotite massifs (Brueckner and Medaris, 2000), as these have commonly experienced subduction-related metamorphism and exhumation.

6. Conclusions

We applied Sr- and Nd-isotope systematics to a well-characterized set of pyroxenites from the Cabo Ortegal Complex, Spain, aiming to provide better constraints on the source and timing of their formation and evolution, which relate to primitive arc magmatism, subduction and exhumation.

Our data show that the different types of Cabo Ortegal pyroxenites studied here do not represent the metamorphic products of low-pressure cumulates and are likely to be cogenetic, and cogenetic with related granulites and garnet-bearing mafic rocks. The

homogeneity of their age-corrected $^{143}\text{Nd}/^{144}\text{Nd}$ and $^{87}\text{Sr}/^{86}\text{Sr}$ indicates that an isotopic equilibrium was initially reached during pyroxenite formation, probably following the late-magmatic migration of residual melts. Isotopic perturbations occurred during their subsequent metamorphic amphibolitization, upon the percolation of hydrous fluids, relatively unradiogenic and LREE-enriched. Sr- and Nd-isotope systematics thus consistently indicate that cpx and amphibole have acquired their disequilibrium over two main episodes, in good agreement with petrological and geochemical constraints. It also adds source constraints, and notably indicates the contribution of enriched mantle components (EM I and/or II), suggesting the involvement of a (potentially old) crustal component, which is consistent with the derivation of Cabo Ortegal pyroxenites and granulites beneath an ensialic island arc.

Sr- and Nd-isotope geochronology consistently provides two main clusters of ages related respectively to the late-magmatic/metamorphic episode accompanying the formation of the Cabo Ortegal pyroxenites and to their metamorphic amphibolitization during syn-subduction exhumation, following the previously suggested arc-root delamination. In good agreement with second-stage Nd model ages, we propose that the Cabo Ortegal pyroxenites record the incipient stage of a Cambro-Ordovician to Neoproterozoic arc (459–762 Ma), reworking a rifted margin of Gondwana, or of a pre-Gondwanan continental block.

Modelling of the timescale of diffusional re-equilibration of REE has also been applied to the same set of pyroxenites based on two-pyroxene REE thermometry and shows an overall consistency with Neoproterozoic ages obtained from radiogenic-isotope geochronology. However, it only provides poor additional constraints on the timing of pyroxenite formation, essentially due to the great sensitivity of this method to grain size and temperature. We thus suggest that the use of model ages corrected for radiogenic ingrowth combined with isochron ages is adequate to constrain the age of subduction- and/or collision-related pyroxenites, considering the potential deformational, metamorphic and metasomatic overprints. This case study of the Cabo Ortegal pyroxenites also illustrates the need for robust petrological and geochemical constraints to interpret the ages of recycled mantle rocks.

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Appendix A. Supplementary material

Supplementary material related to this article can be found online at <http://dx.doi.org/10.1016/j.epsl.2017.07.017>.

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